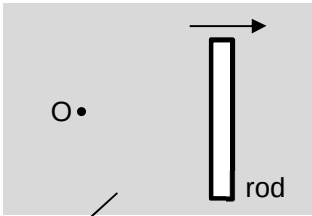
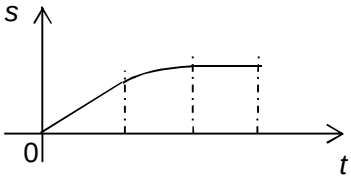
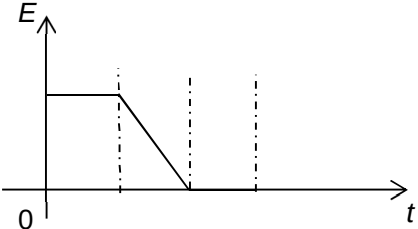
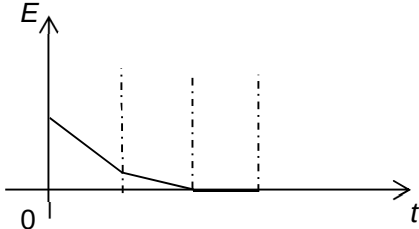
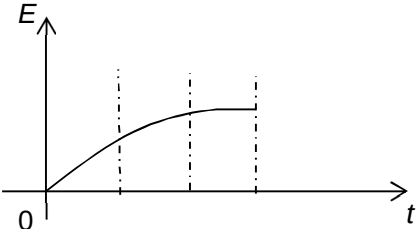
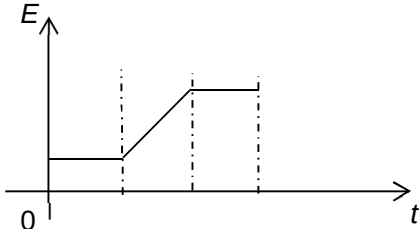


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Diagram A shows an aluminium rod moving away from point O, at right angles to a uniform magnetic field. Diagram B shows the variation with time t of its displacement s from O.

	 region of magnetic field Diagram A	 Diagram B
	A 	B 
	C 	D 

Answer: A

By Faraday's law, e.m.f. induced is the rate of change of magnetic flux linkage. Since the rate of change of magnetic flux linkage is proportional to the velocity of the rod, the e.m.f.-time graph has the same shape as the velocity-time graph. i.e. refers to the gradient of s-t graph.